

CSC633 MACHINE LEARNING



TEXAS A&M UNIVERSITY

Department of Computer
Science & Engineering

CSCE 633: Machine Learning

Logistic Regression

Goals

- What is Classification?
- Why not use Linear Regression?
- Logistic Regression
- Odds
- Maximum Likelihood Estimation
- Logistic Regression optimization
- Gradient Descent and Logistic Regression
- Misc. Logistic Regression

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- $p(x) = 0.9$ and $\frac{0.9}{0.1} = 9$ - with odds of 9.

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$$\hat{\theta} = \operatorname{argmax}_{\theta} \mathcal{L}(\theta; x)$$

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- Maximizing this means finding w_0 and w_1 that maximizes this product

Multiple Logistic Regression

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- $NLL(w) = \sum_{i=1}^n \log(1 + e^{-\tilde{y}_i w^T x_i})$ which has no closed form solution.

Takeaways

- Odds
- Maximum Likelihood Estimation